

Representation Repair under Distribution Shift in a Partially Observable Card Game

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(Dated: August 23, 2026)

Sequential policies with variable legal-action sets can omit which observed item a legal action selects even when positional and other action attributes remain available. We study such an identity-information defect in a partially observable stochastic card-game simulator. The historical encoder represented ordinary PLAY options without binding each option to the observed hand card/template ID (the source-card type ID) that it selected. The model therefore saw pooled hand card IDs and an option’s hand index, but not their relational binding. We repaired that binding and fine-tuned four output-side modules of a frozen baseline on recent, winner-only replay-derived decisions. A KL term used frozen A2 weights evaluated under the cell’s encoder. We compared the resulting policy, EXP23, with the frozen control, C0, on an engine-seed-, order-, and seat-matched schedule prospectively specified in a locally committed protocol before result access, over seven fixed opponent policies and both actual turn orders. In that one execution, the observed equal-weight schedule-paired win-rate difference was 2.786 percentage points (95% conditional empirical interval from stratified paired resampling [0.679, 4.929]; 2,800 pairs). The interval is conditional on the realized execution and omits run-to-run variation from two opponents’ wall-clock-bounded search. The secondary fixed-group point estimates were 3.88 percentage points for the defined matched-policy family and 1.33 percentage points for the other three opponents. In a separate episode-clustered reanalysis of retained, outcome-selected replay units, EXP23 approval among binary candidate-control disagreements was 0.455 (95% interval [0.353, 0.567]); The interval included one half. A planned seven-opponent four-cell analysis was invalidated because repeated control outcomes differed on 210 of 2,800 seed-condition units, so its cross-cell mechanistic contrasts are not estimable. Two retained null experiments further bound broader claims. The study separates representation validity, imitation agreement, paired gameplay performance, and opponent-group heterogeneity; it does not claim universal improvement or superhuman play.

I. INTRODUCTION

Imperfect-information games are useful environments for studying sequential decision making because an agent must act from a partial view of a stochastic process while anticipating another policy. Poker systems such as Deep-Stack and Pluribus illustrate the scientific difficulty of this class [1, 2]; they are context, not empirical comparators for the card-game agent studied here. Our question is narrower: what happens when a concrete legal-action representation defect is repaired, and do offline imitation metrics rank the repaired policy in the same way as paired simulated outcomes?

The distinction matters because a representation can be more faithful to the available observation without increasing agreement with outcome-selected, held-out recorded actions. Conversely, a policy can improve terminal outcomes against a defined opponent set without reproducing more historical actions. Sequential imitation learning is exposed to distribution shift because a learned policy changes the states it visits [3]. The present study does not implement DAgger and claims none of its guarantees; that work motivates separating held-out action agreement from on-policy outcomes because learner-induced state distributions can differ.

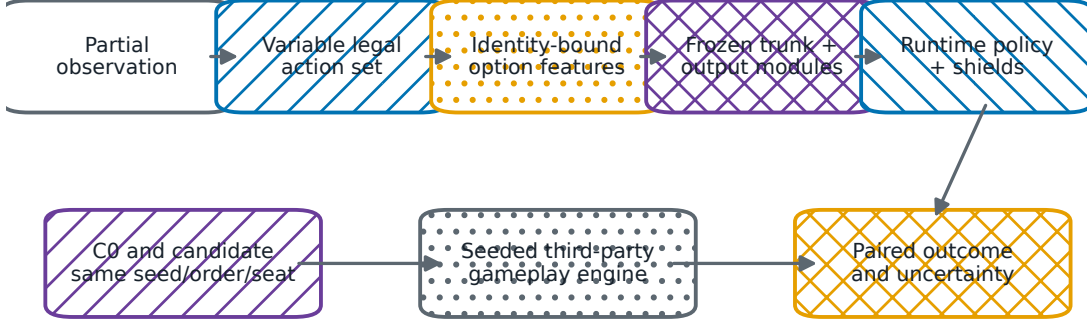
Legal-action interfaces add a separate representational

hazard. Invalid-action masking restricts selection to valid actions [4]; that legality mechanism is separate from whether features encode each valid option’s semantic attributes. Permutation-invariant and permutation-equivariant set architectures provide a vocabulary for variable-size collections [5]. Here the defect was bounded: an ordinary PLAY option carried an index into the observed hand, but the v2 feature path left its source-card-ID field at zero. The policy retained pooled state tokens for hand card IDs, the normalized hand index, and other option attributes, but lacked the mapping between index and identity. Omission of that relation does not by itself imply that two simultaneous option vectors were identical.

We contribute (i) a corpus-level audit of this identity omission; (ii) an identity-aware repair coupled to a restricted, A2-anchored output-module update; (iii) a fresh, engine-seed schedule-paired, order-balanced gameplay execution; (iv) an audit showing why the planned four-cell representation-by-training analysis failed its repeated-control parity requirement; and (v) an explicit comparison with episode-clustered recorded-action agreement and retained null experiments. Reporting was informed by recommendations to quantify aggregate uncertainty and improve reproducibility through transparent reporting and artifact sharing [6, 7]. Our hash-based provenance scheme is a study-specific implementation. The opponents are a frozen engineering population rather than a probability sample. Public competition rating and rank are not outcomes in this paper.

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Policy and matched evaluation pipeline



Engine seed, opponent, order, and seat define schedule pairs; opponent search may vary with runtime.

FIG. 1. Policy and evaluation pipeline. Candidate and control use the same opponent, actual order, physical seat, and engine seed within each pair. This schedule match does not couple wall-clock-bounded opponent search decisions. The restricted engine and opponent implementations are fixed inputs rather than redistributable contributions. OpenAI Codex assisted with plotting-code drafting; source inputs and the rendered figure were checked against the cited artifacts.

II. FORMAL PROBLEM SETTING

We model the simulator as a finite-horizon partially observable stochastic game. At engine step t , the acting agent receives observation o_t and a variable legal-action collection $A_t = \{a_{t1}, \dots, a_{tK_t}\}$. The observation contains only information exposed by the competition interface. A policy scores each legal action,

$$s_\theta(o_t, a_{tj}) = h_\theta(g_\theta(o_t), \phi_\theta(o_t, a_{tj})), \quad (1)$$

and selects from A_t subject to deterministic runtime constraints. Here g_θ represents global observation features and ϕ_θ represents one legal option. Because K_t varies, option identity is part of the policy contract rather than merely an output label.

For a fixed observation, define an exact blind-input collision when distinct legal actions $a \neq a'$ have the same complete option-head input after the source-card-ID coordinate is set to zero. A weaker, component-level omission occurs whenever that coordinate is zero although the selected observed card has a known type ID. The defect examined here is the latter: ordinary PLAY options refer to entries of the observed hand, the historical encoder assigned source ID zero, and the repair binds the in-range selected card ID. This ID is not a unique physical-instance serial. Other features, including normalized hand index, can still distinguish two options. The v2 state embedding pools visible hand-identity tokens, while the local option vector carries the hand index; neither encodes which identity occupies that index. We therefore describe the defect as a missing option-item binding and measure exact within-state collisions separately rather than assuming them from the missing coordinate.

For paired evaluation, let $W_{ci}^{(1)}$ and $W_{ci}^{(0)}$ be EXP23 and C0 win indicators for paired unit i in opponent-by-order cell c . Losses and draws are zero for the primary endpoint. Define

$$D_{ci} = W_{ci}^{(1)} - W_{ci}^{(0)}, \quad \hat{\Delta} = \frac{1}{14} \sum_{c=1}^{14} \frac{1}{n_c} \sum_{i=1}^{n_c} D_{ci}. \quad (2)$$

Every n_c equals 200, so $\hat{\Delta}$ also equals the mean over all 2,800 paired units after complete execution. The estimand is finite-population: it concerns the seven frozen opponents, two actual-order conditions, policy packages, and engine build named in the protocol.

The replay estimand is different. At an outcome-selected recorded decision where EXP23 and C0 select semantically different actions, the recorded action may approve EXP23, approve C0, or approve neither. Approval is

$$\hat{A} = \frac{N_{\text{EXP23}}}{N_{\text{EXP23}} + N_{\text{C0}}}, \quad (3)$$

with neither-approved cases reported as abstentions. Episodes, not decisions, are the uncertainty clusters because within-episode decisions are dependent [8]. This conditional quantity is not gameplay value.

III. ENVIRONMENT AND FROZEN BASELINE

The environment is a two-player, partially observable card-game simulator supplied for a competition. The engine exposes observations, legal actions, and actual turn order. Its rules, binaries, card database, deck material, and third-party opponent packages are neither claimed as

contributions nor included in the sanitized release. End-to-end engine replication requires separately authorized access.

The frozen control c0 is the A2 neural policy with a deterministic Damage V0 runtime layer. Both actual-order arms load byte-identical A2 weights (SHA-256 b19871a9...b6bda8); the control package tree is 13426288...535c3. Damage V0 is post-policy logic, not a separately trained damage model. EXP23 enables the identity-aware encoder and loads byte-identical updated model arrays in all three packaged policy files (model cefe6118...96984, tree 83489e0c...7f1c0, archive 0734b60c...e7096). Full digests and hash algorithms are in the companion repository documentation and claim ledger.

One archived EXP23 order-policy manifest incorrectly names the A2 model hash, and one historical certificate reports a stale candidate tree hash. Direct archive-member bytes, the tracked frozen-hash inventory, and every evaluation record agree on the hashes above. Following the prespecified evidence hierarchy, bytes and raw evaluation records are authoritative; the inconsistent metadata are preserved and disclosed.

IV. REPRESENTATION DEFECT AND IDENTITY-AWARE INTERVENTION

The historical v2 branch resolved ordinary PLAY options with no explicit board area to no source object. The feature serializer then emitted `source_card=0`. The repair is deliberately narrow: with its package flag enabled, an in-range legal-option index resolves `hand[option.index]` and serializes that observed item's card/template ID (the source-card type ID). It does not inspect hidden state. A parity audit over 4,659 decisions and 24,065 options found 2,853 changed PLAY source values across 658 decisions, all in the main-action context, and no other feature changes.

The retained corpus contains 47,653 decisions and 37,199 ordinary PLAY options. Reconstructing the blind encoder left 37,199 of those options unresolved; the repaired rows bound all of them. The retained hand-position coordinate matched the raw index for all 37,199 instances, and all 10,522 states with multiple PLAY options had distinct indices. Of 15,728 states containing PLAY, 9,776 (62.2%) exposed at least two distinct PLAY card IDs. An exact within-state audit of the complete blind option-head input found 0 cross-identity collision groups in 0 states (0.0% of PLAY-option instances). Thus the retained rows verify pervasive identity-coordinate omission, not exact full-vector aliasing. All affected options occurred in the main-action context. Source-frequency outputs use anonymous tokens so no proprietary card names or identifiers enter the release. Because the exact collision subgroup contained no rows, its output-disagreement and recorded-action approval rates are not estimable; the machine-readable au-

dit records zero counts and null rates explicitly, while the noncollision group equals the full corpus.

Applying the two policy models directly to retained feature rows produced an overall index-exact disagreement rate of 13.7%. The rate was 44.4% in multi-identity states and 5.8% elsewhere (Table I). This locates model-output sensitivity; it omits production runtime constraints and is not a gameplay or causal subgroup estimate.

V. REPLAY DATA AND RESTRICTED OUTPUT-MODULE TRAINING

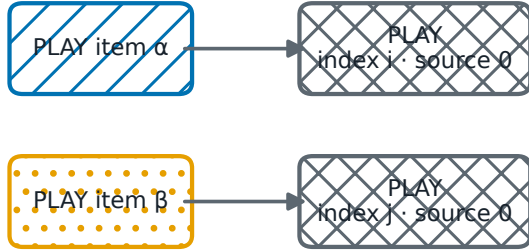
The retained feature corpus contains 47,653 decisions: 38,254 training rows, 3,361 internal-validation rows, and 6,038 rows from six held-out teams. It spans 472 episode identifiers and 69 team labels. Every retained row is labeled with positive reward and the historical `daily_top_episode` source, so the corpus is outcome-selected rather than representative of play; demonstrator expertise and recorded-action optimality were not independently established. The split manifest gives held-out-team precedence and records no duplicate keys. A team is "held out" only from this refresh's gradients, internal validation, and checkpoint selection; A2's original training corpus is unavailable, so the teams are not proven unseen to the initialized model. A nominal temporal holdout is invalid: its eight files are empty stubs and supply no rows. All retained rows have a null observation field, and the original Aug. 14–15 replay observations used to mine the corpus were deleted by the historical pipeline. Thus the retained feature rows—not end-to-end replay extraction or re-resolution of `hand[option.index]`—are the reproducible training source. The separate parity audit tests the binding on observations. The accompanying data card documents intended use and privacy boundaries in the spirit of dataset documentation practice [9].

Training initialized student and teacher from A2, froze all parameters outside the option projection plus score, count, and value heads, and optimized the student on CPU for three epochs with AdamW. The frozen A2-weight teacher was evaluated under the same encoder as its cell; for the identity-aware cell this is not identical to blind c0 behavior. The learning rate was 10^{-4} , weight decay 10^{-5} , batch size 256, seed 20260816, gradient-norm limit 1.0, and KL distillation weight 0.5. Eight of 19 exported arrays changed, exactly those of the four trainable modules; frozen arrays were byte-preserved.

The configured fresh-data weight was 0.999. Historical batch rounding made the implied rehearsal target zero: each epoch processed 38,254 fresh and zero rehearsal rows. We therefore use *conservative* only for frozen parameter scope, baseline initialization, and KL anchoring. We do not describe the realized update as a 99.9/0.1 mixture or claim a formal safe-improvement guarantee. Baseline-anchored policy improvement provides related motivation, not a theorem for this update [10].

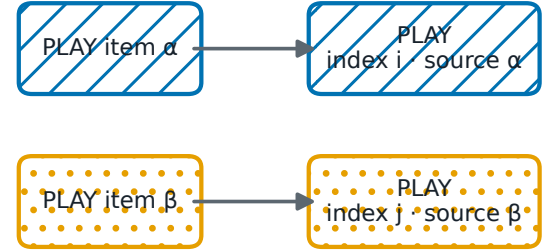
Option-item binding repair in the variable legal-action encoder

Before: item binding omitted



37,199 / 37,199 ordinary PLAY options
lack source binding

After: source identity retained



9,776 states expose ≥ 2 distinct
PLAY source-card IDs

FIG. 2. The bounded encoding defect and repair. Before repair, each ordinary PLAY option had source identity zero while retaining its position and other attributes. After repair, the in-range hand index also binds source identity. The schematic does not assert full-vector collisions; symbols are abstract and do not depict game assets. OpenAI Codex assisted with plotting-code drafting; source inputs and the rendered figure were checked against the cited artifacts.

TABLE I. Representation and output-module diagnostics on outcome-selected retained feature/action rows. These are not gameplay outcomes.

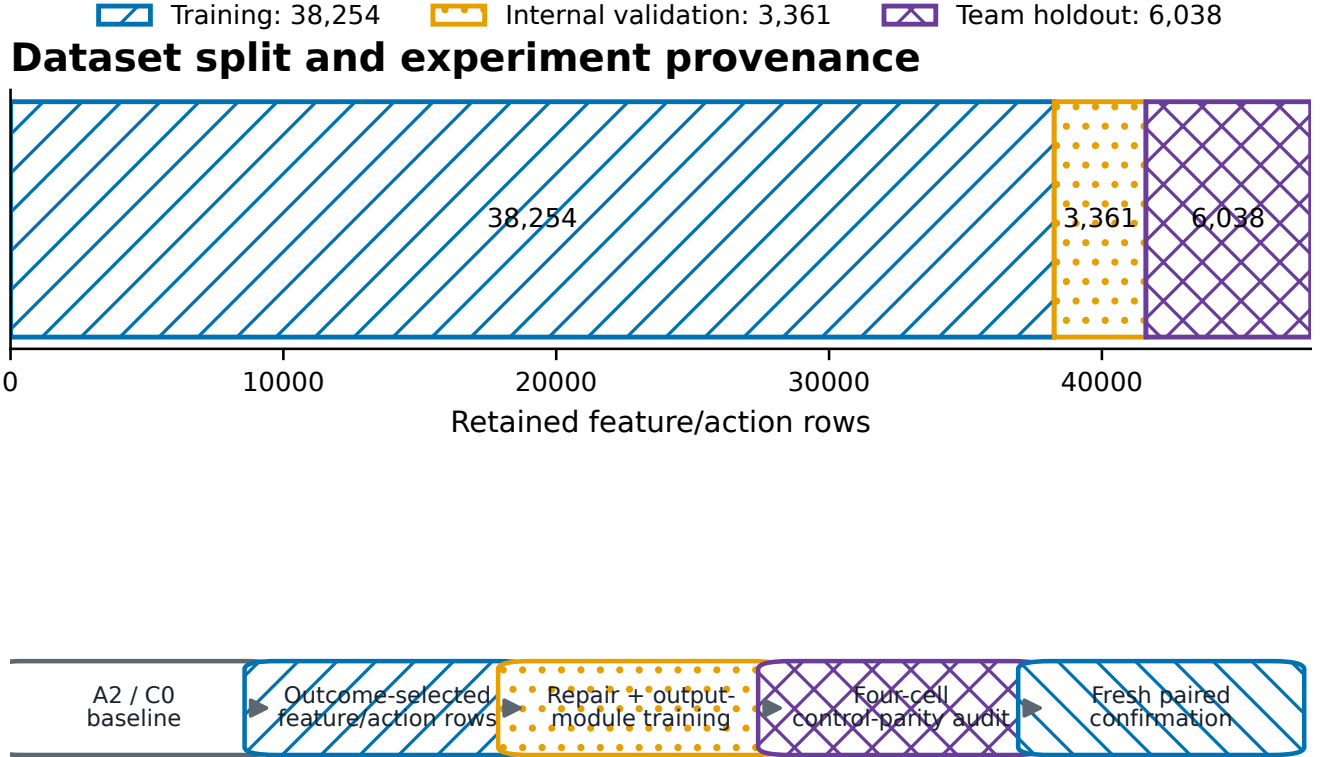
Diagnostic	Value
Decisions	47,653
Ordinary PLAY options	37,199
Blind unresolved / repaired bound	37,199 / 37,199
Multi-identity PLAY states	9,776 (62.2%)
Index coordinate matches raw / all multi-PLAY unique	37,199 / 10,522
Exact within-state blind-input collisions	0
Head disagreement, multi-identity	44.4%
Head disagreement, other	5.8%

VI. EVALUATION PROTOCOL AND STATISTICAL ANALYSIS

The fresh protocol was prospectively committed in the local repository before result access. It freezes C0, EXP23, the seeded engine, production-engine integrity sentinel, seven opponent package hashes, actual-first and actual-second conditions, disjoint seed ranges, physical-seat alternation, and 200 candidate-control pairs in every cell. Within a pair, candidate and control records share opponent, engine seed, actual order, and physical seat. The frozen command also specifies the opponent environment and maximum-decision limit, but the per-game rows do not serialize either field; those two settings therefore cannot be independently verified row by row.

The evaluator dispatches candidate and control games as separate process-pool tasks, and native opponent-search state is process-local rather than explicitly reset or coupled by the schedule. A post-execution source audit found that the Starmie opponent stops search at a `time.monotonic()` wall-clock deadline and the Dipplin opponent likewise uses soft and hard monotonic-clock deadlines. Thus the schedule couples the seeded gameplay engine but does not couple all opponent decisions or scheduler timing. The design intended the variance reduction associated with common random numbers [11], but the implementation does not establish a full common-random-number coupling and we claim no such precision gain.

The complete schedule comprises 2,800 paired units



Raw replay extraction cannot be rerun: original Aug. 14–15 observations are missing.

FIG. 3. Retained data split and experimental provenance. Counts come from the frozen manifest. The missing source replays limit feature-extraction reproducibility but do not alter the retained-row transform used to train the C3 diagnostic. OpenAI Codex assisted with plotting-code drafting; source inputs and the rendered figure were checked against the cited artifacts.

and 5,600 engine games. The binary primary outcome maps a win to one and loss or draw to zero. The secondary utility maps win, draw, and loss to 1, 0, −1. An exception, nonterminal truncation, illegal action, candidate/control/opponent policy error, incomplete pair, hash drift, production-sentinel change, or pairing mismatch invalidates a cell rather than being deleted or imputed. All scheduled fresh pairs completed with zero recorded policy errors. The frozen runner did not record per-game latency; parallel wall time is not substituted. The primary result is therefore the observed prespecified schedule-paired contrast from one execution, not a claim that repeated executions are invariant.

The primary interval uses 100,000 bootstrap replicates. Each replicate resamples complete candidate-control pairs with replacement independently within all 14 cells, then averages the cell means. The exact two-sided McNemar test conditions on candidate-only plus control-only wins and tests symmetry of the discordant direction. This is sharper than the weak null of an equal mean across 14 fixed strata, for which the paired bootstrap interval is the primary numerical summary. It

conditions on the realized execution and does not include run-to-run scheduler or opponent-search variation. Fourteen individual cell tests are secondary and Holm-adjusted together. Opponent, order, family, and utility summaries remain descriptive. No meta-weighted estimate is computed because the historical frozen weight file is absent.

The planned four-cell ablation fixed the same engine, opponents, seed schedule, orders, seats, and pair counts across separately executed runs. It embeds the already reported C4–C1 primary execution and adds C2 and C3 runs, each with its own repeated C1 arm; it is not a second independent C4 confirmation. C1 is blind encoder plus A2; C2 is identity-aware encoder plus unchanged A2; C3 is blind encoder plus matched output-module retraining; and C4 is identity-aware encoder plus the matched update (EXP23). C2 and C3 are mechanistic probes, not deployment candidates. The prospectively frozen analyzer gate required the repeated C1 outcomes to agree for every seed-condition unit. That gate failed on 210 of 2,800 units: 191 against Starmie and 19 against Dipplin. Accordingly, the planned seven-opponent four-cell con-

trasts and their inferential procedures are not estimable under the protocol. In particular, the C4–C2, C4–C3, and difference-in-differences comparisons require cross-execution parity that failed. Realized within-run C2–C1 and C3–C1 summaries are retained only as process-sensitive descriptions, and any five-opponent available-record-parity subset is a post hoc exploratory diagnostic. The C3 transform zeroed exactly 37,199 source fields in 47,653 retained rows, completed all three epochs, realized zero rehearsal rows, and exported model hash `d22d0b52...8b70`.

a. AI-assisted methods. OpenAI Codex, using a GPT-5-family model whose exact deployed snapshot was not exposed, assisted under human direction with analysis-code drafting, statistical checks, visualization code, artifact auditing, and manuscript drafting. Its outputs were not treated as evidence: reported quantities were regenerated from retained artifacts and checked by executable validation, source inspection, hash comparison, and figure review. Compilation, full-page rendering, and visual inspection of the final PDF are required before submission. Further details and the limitation on unrecoverable earlier AI use appear in the disclosure below.

VII. PRIMARY GAMEPLAY RESULTS

Across the frozen population, EXP23 produced 1763 wins and C0 produced 1685. These marginal counts do not replace the schedule-paired analysis because both policies may win or fail to win in the same pair. The observed schedule-paired win-rate difference was 2.786 percentage points (95% interval [0.679, 4.929]). Discordant outcomes included 496 EXP23-only and 418 C0-only wins; the exact two-sided McNemar value was 0.0108. That test addresses the sharper symmetry null for discordant directions and is secondary to the stratified interval. The interval excludes zero for this realized execution, but it does not quantify between-execution timing variation. Candidate/control draw totals were 1/2. The secondary win/draw/loss utility effect was 0.055 (95% interval [0.013, 0.098]).

Actual-first and actual-second effects were 3.14 percentage points (95% interval [0.21, 6.07]) and 2.43 percentage points ([-0.64, 5.50]), respectively. The defined matched-policy aggregate was 3.88 percentage points ([0.88, 6.88]); the other three frozen opponents yielded 1.33 percentage points ([-1.58, 4.17]). These are fixed-group summaries, not estimates for unseen opponents or policy families.

VIII. INVALIDATED FOUR-CELL REPRESENTATION-BY-TRAINING ANALYSIS

All scheduled C2 and C3 games completed, but the planned four-cell analysis failed its prospectively frozen analyzer gate before contrast estimation (Table III). For

each seed-condition unit, that analyzer required the C1 outcome repeated alongside C2, C3, and C4 to be identical. At least one of the three C1 outcomes differed on 210 of 2,800 units: 191 against Starmie and 19 against Dipplin. The other five opponents had identical available win/draw/error/decision-count summaries across their repeated C1 runs; traces were not captured. Source inspection showed wall-clock search limits in the two discordant opponent implementations, while the evaluator ran each game in a separate process-pool task. Equal engine seeds therefore did not define equal realized opponent behavior.

All planned seven-opponent four-cell contrasts and their inferential procedures are consequently invalidated; no confirmatory mechanistic attribution is made. The cross-execution C4–C2 and C4–C3 comparisons and difference-in-differences interaction are not estimable under the frozen protocol. The C4–C1 number remains the observed primary within-execution comparison reported above, but the parity failure limits its uncertainty interpretation. C2 versus its own concurrently executed C1 arm and C3 versus its own C1 arm are retained only as all-seven process-sensitive descriptions. A post hoc five-opponent subset, restricted to opponents whose available C1 summaries happened to agree, is labeled exploratory and is not a replacement for the invalidated analysis.

IX. REFRESH-HELD-OUT RECORDED-ACTION DIAGNOSTICS ACROSS DIFFERENT STATE DISTRIBUTIONS

Two secondary, exploratory offline diagnostics address different layers; their records had been inspected in historical certification work and are newly recomputed here, not newly unseen. First, direct model scoring on 6,038 held-out-team feature rows produced 821 index-exact model disagreements. Of 579 binary decisive cases, EXP23 matched the recorded action 271 times and C0 308 times; 242 additional disagreements were abstentions. EXP23 approval was 0.468 (episode-bootstrap 95% interval [0.423, 0.513]). The interval included one half. This is a model-only diagnostic without runtime constraints. Its recorded actions come from winner-only top-episode rows and are not independently certified as optimal.

Second, we ran the unchanged historical semantic-action evaluator on retained 2026-08-13 replay units that were not part of the missing Aug. 14–15 source set. Eligibility was restricted to winning units from five prespecified teams using the exact target deck. Among 169 semantic disagreements, 60 approved EXP23, 72 approved C0, and 37 were abstentions. The 132 binary decisions across 16 episodes yielded approval 0.455 (95% interval [0.353, 0.567]); the equal-team mean was 0.492 across the 2 teams with eligible units (three prespecified teams contributed none). The interval included one half. Table IV separates multi-identity and other decision states. Both diagnostics are conditional on their fixed, previ-

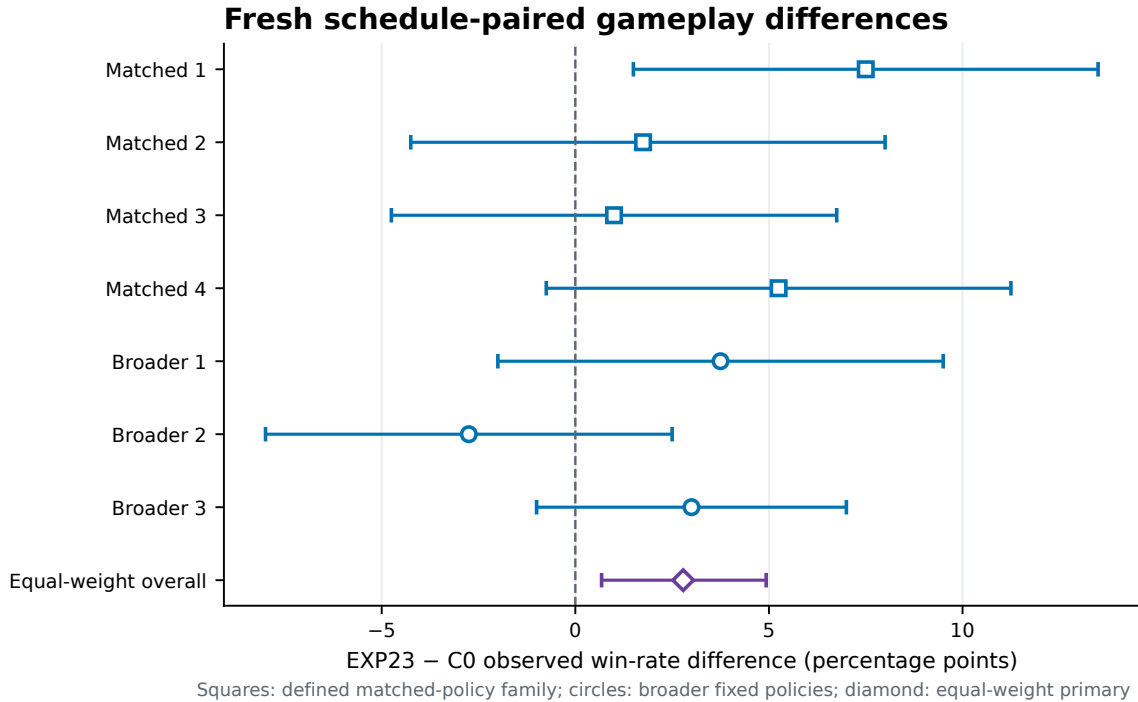


FIG. 4. Fresh schedule-paired effects by opponent, averaged over actual order, and the prospectively specified equal-weight primary effect. Intervals resample observed pairs within cells and condition on the single realized execution. Opponents are a fixed population; individual points are secondary. OpenAI Codex assisted with plotting-code drafting; source inputs and the rendered figure were checked against the cited artifacts.

TABLE II. Fresh, prospectively specified schedule-paired comparison by fixed opponent and actual order. Effects and conditional empirical percentile intervals are percentage-point differences in observed win rate (EXP23 minus C0).

Opponent	Order	Pairs	Effect	95% CI	Holm p
Matched 1	first	200	+7.0	[-1.5, +15.0]	1.000
Matched 1	second	200	+8.0	[-0.5, +16.5]	1.000
Broader 3	first	200	+4.5	[-0.5, +9.5]	1.000
Broader 3	second	200	+1.5	[-4.5, +7.5]	1.000
Matched 2	first	200	+7.0	[-1.5, +15.5]	1.000
Matched 2	second	200	-3.5	[-12.0, +5.0]	1.000
Broader 2	first	200	-4.0	[-11.5, +3.5]	1.000
Broader 2	second	200	-1.5	[-9.0, +5.5]	1.000
Matched 3	first	200	-5.0	[-13.5, +3.5]	1.000
Matched 3	second	200	+7.0	[-1.0, +15.0]	1.000
Matched 4	first	200	+7.0	[-1.5, +15.5]	1.000
Matched 4	second	200	+3.5	[-5.0, +12.0]	1.000
Broader 1	first	200	+5.5	[-1.5, +12.5]	1.000
Broader 1	second	200	+2.0	[-7.0, +11.0]	1.000

ously inspected team sets and selection rules; their intervals represent episode variation within those sets, not between-team sampling uncertainty or performance for unseen teams.

The mismatch is substantively plausible. Replay approval asks which policy reproduces one historical action

in states generated by historical agents. It does not reward a different action for downstream value. Paired gameplay instead compares terminal outcomes on trajectories induced by each policy. The estimands differ in state distribution, conditioning, horizon, and analysis unit. Their disagreement is consistent with metric

TABLE III. Fail-closed four-cell control-parity audit and post-failure descriptive sensitivities. The planned seven-opponent cross-cell analysis was not estimable because repeated C1 records disagreed. Within-run effects are process-sensitive realized comparisons. The five-opponent subset was selected post hoc by whole-package serialized-control parity; it changes the target population and has no inferential interval or test.

Analysis	Contrast/audit	Pairs	Effect (pp)	Status
Planned seven-opponent	C1 parity mismatches	2,800	–	invalidated (210 units)
Within own run	C2-C1 (C2 run)	2,800	-0.107	descriptive only
Within own run	C3-C1 (C3 run)	2,800	-0.679	descriptive only
Within own run	C4-C1 (C4 run)	2,800	+2.786	descriptive only
Post hoc five-opponent	C2-C1	2,000	-1.40	descriptive only
Post hoc five-opponent	C3-C1	2,000	-0.70	descriptive only
Post hoc five-opponent	C4-C1	2,000	+3.70	descriptive only
Post hoc five-opponent	C4-C2	2,000	+5.10	descriptive only
Post hoc five-opponent	C4-C3	2,000	+4.40	descriptive only
Post hoc five-opponent	Interaction	2,000	+5.80	descriptive only

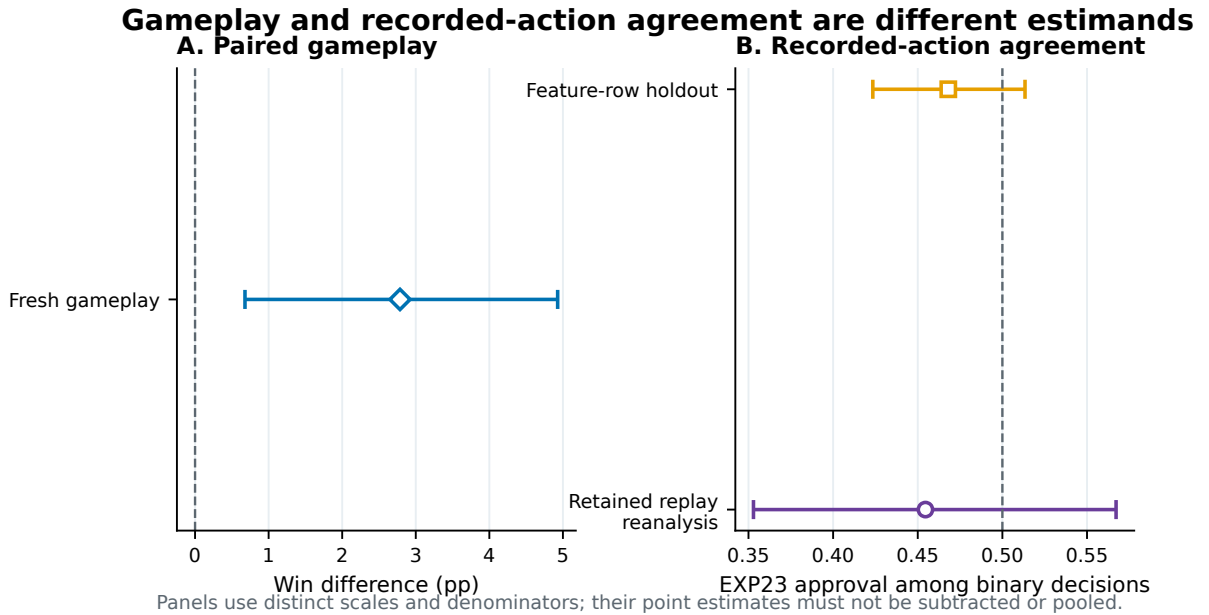


FIG. 5. Gameplay and recorded-action agreement are different estimands and use different axes. Panel A shows the paired terminal win difference. Panel B shows EXP23 approval conditional on binary policy disagreement; neither-approved cases are abstentions. The panels must not be numerically pooled or subtracted. OpenAI Codex assisted with plotting-code drafting; source inputs and the rendered figure were checked against the cited artifacts.

dependence across different state distributions; the two estimates cannot be directly tested against or subtracted from one another. It does not establish that outcome-selected recorded-action data are poor or that simulated gameplay is unbiased.

X. NEGATIVE AND NULL EXPERIMENTS

Two bounded experiments retain raw artifacts sufficient for independent reaggregation (Table V). An actual-second two-turn temporal takeover produced 0.083 per-

centage points over 1,200 paired units across three fixed internal opponents, with an opponent-stratified paired bootstrap interval $[-2.000, 2.167]$. The interval spans zero, so that confirmation did not demonstrate benefit. A preceding positive selection screen was not pooled with confirmation, and the result does not show that temporal policies fail generally.

A bounded second-own-turn sequence oracle evaluated 960 worlds in 60 root clusters against three fixed opponents, permitting at most two semantic deviations. Its effect was 0.313 percentage points with root-cluster interval $[-1.458, 1.979]$, below a historically reported $+3$ -

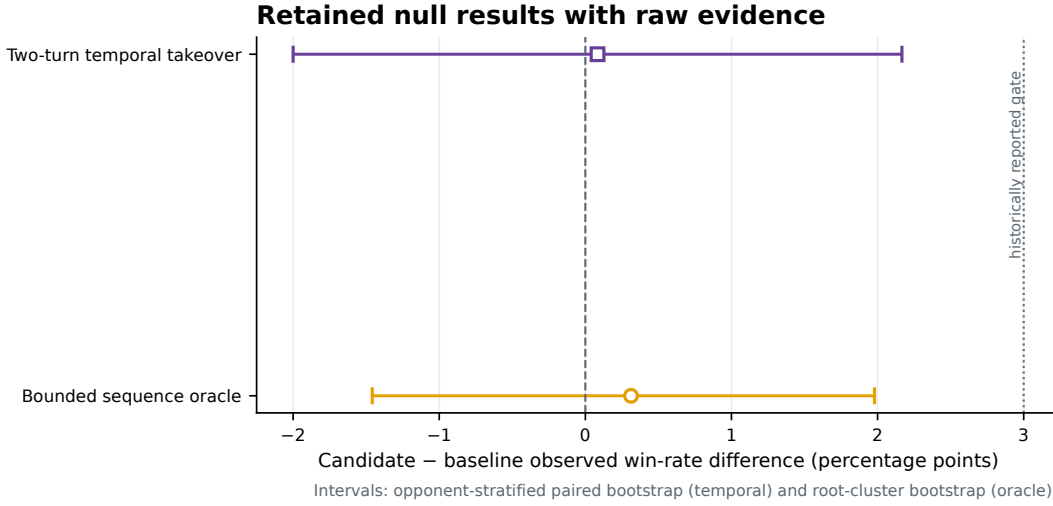


FIG. 6. Retained null results with surviving raw evidence. The temporal interval resamples paired units within opponent; the oracle interval resamples root clusters. PPO, expected-Q, and the older Turn Director are absent because their underlying result rows do not survive. OpenAI Codex assisted with plotting-code drafting; source inputs and the rendered figure were checked against the cited artifacts.

TABLE IV. Recorded-action agreement conditional on EXP23–C0 semantic disagreement in refresh-held-out, previously inspected 2026-08-13 replay units. Abstentions are excluded from the approval denominator.

Stratum	EXP23	C0	Abstain	Approval
Overall	60	72	37	0.455
multi play identity	38	45	32	0.458
other	22	27	5	0.449

TABLE V. Independently reaggreated null results with surviving raw artifacts.

Experiment	Units	Effect	95% CI
Two-turn temporal takeover	1,200	+0.083	[-2.000, +2.167]
Bounded sequence oracle	960	+0.313	[-1.458, +1.979]

point gate whose advance specification cannot be independently established from reachable Git history. This constrains one proposal-and-confirmation procedure; it is not a test of planning as a class.

Project audits also describe a shielded actor-only PPO update, an expected-Q labeling exercise, and an older Turn Director. Their exact result rows or packages are absent, so their reported numerical estimates are excluded from the evidentiary manuscript. They are retained as summary-only provenance entries rather than silently promoted to raw evidence or generalized as failures of reinforcement learning and search.

XI. DISCUSSION

The case separates four quantities that are often collapsed: representational validity, held-out imitation agreement, paired gameplay value, and opponent-group heterogeneity within the frozen battery. Binding a legal action to the observed item it selects repairs a factual defect. It does not imply that inherited embeddings are calibrated, that every downstream choice improves, or that the recorded outcome-selected action is uniquely valuable. The failed four-cell parity gate also shows that matching engine seeds is not enough to distinguish encoding, training, and their interaction when opponent search depends on process timing. The observed C4–C1 difference therefore cannot be assigned entirely to the encoder repair or to retraining.

The restricted update is a practical audit pattern for a competent baseline. Freezing most parameters and retaining a KL term to frozen A2 weights under the cell’s encoder reduces the dimensions in which learning can move. It does not provide a monotonic policy improvement guarantee, particularly because the realized stream contained no rehearsal rows and because replay and gameplay state distributions differ.

An engine-seed schedule match supports pairing on recorded design variables, but the audit demonstrates that it did not couple two opponents’ time-bounded search paths. The primary estimate and interval summarize one realized execution; they do not establish common-random-number variance reduction or repeated-run stability. Opponent packages, actual order, engine build, process scheduling, runtime constraints, and decks remain part of the estimand. Similarly, a clustered replay interval respects episode dependence but cannot turn a

conditional imitation proxy into long-horizon value.

The negative results and evidence conflicts narrow the story. Additional temporal or bounded sequence structure did not demonstrate benefit in their bounded tests. Historical favorable aggregate claims were not used when their rows, weights, or estimator could not be reconstructed. This reduces apparent evidence but provides a direct, auditable path from every included number to retained bytes.

XII. LIMITATIONS

First, the seven opponents are a deliberately frozen convenience population, several with related development histories. They are neither random nor exhaustive. Bootstrap intervals resample observed seed-condition pairs under this battery and do not support population-wide generalization.

Second, engine seed, actual order, physical seat, and opponent package were matched within each scheduled pair, but candidate and control games ran as separate process-pool tasks. Starmie and Dipplin use wall-clock-bounded search, so their decisions can vary with scheduling. Repeated C1 outcomes disagreed on 210 of 2,800 units across ablation executions. The primary interval conditions on a single execution and omits this run-to-run component; the planned seven-opponent four-cell cross-run analysis was invalidated.

Third, only one training seed and one outcome-selected retained corpus were evaluated. All corpus rows came from winning top-episode units, and the replay reanalysis further required a prespecified team and exact target deck. These selection rules limit both imitation and held-out estimands; the replay intervals are conditional on the fixed team sets and include no between-team population uncertainty. C2 supplies a previously zero feature to weights not trained for it, C3 and C4 were separately fitted, and only one training seed was used. Within-run C2–C1 and C3–C1 summaries are process-sensitive diagnostics, not valid substitutes for the failed cross-run design. Model-only disagreement omits runtime constraints, and multi-identity-state associations do not identify a causal subgroup effect.

Fourth, original replay observations for the retained Aug. 14–15 feature corpus and an original environment lock are missing. The nominal temporal holdout is empty. Training can be repeated from retained feature rows, but feature mining cannot be reconstructed from source replays. The fresh runner did not store per-game latency.

Fifth, the third-party engine, source, binaries, card data, decks, private replays, and opponent implementations cannot be redistributed under the available terms and privacy constraints. The intended private-review release is designed to recalculate primary marginal counts and the equal-weight point estimate from released paired outcomes, and to regenerate manuscript tables and figures from released processed outcomes and machine-

readable summaries. It does not independently recompute every interval or statistic from engine trajectories. Rights review is also required before redistributing policy weights or placing a permissive license on the companion package.

Sixth, historical seven-policy and meta-weighted aggregates lack their raw wave files and frozen weight table. Historical CERT-B prose calls 529 disagreements “decisive,” although its reconstructed binary denominator is 398, and its episode rows are absent. These values are recorded as conflicts or unsupported and omitted from the main evidence.

Finally, author identity, affiliations, corresponding-author ORCID, CRediT roles, funding, conflicts, acknowledgments, complete historical AI-tool use, archive DOI, and release ownership are not established by repository evidence. They require human confirmation before submission.

XIII. CONCLUSION

Distinct observed card plays were legal, yet their selected card/template IDs were omitted from one historical option-feature coordinate. We repaired that binding, applied a restricted A2-anchored output-module update, and evaluated the result with one engine-seed schedule-paired simulation execution and episode-clustered replay diagnostics. The observed gameplay interval excluded zero conditional on that execution, while the offline approval interval did not establish improved recorded-action agreement. The invalidated ablation shows that seed matching cannot substitute for eliminating or modeling time-dependent opponent behavior. The broader methodological recommendation is to audit legal-action identity explicitly, verify every source of stochastic coupling, pair evaluations at their natural unit, and preserve failed validation gates, negative results, and provenance conflicts alongside favorable findings.

ACKNOWLEDGMENTS

Acknowledgment of the competition organizer, engine provider, data contributors, compute providers, and human reviewers requires permission and author confirmation. Funding and grant identifiers have not been established from the repository and must be supplied or explicitly declared absent.

Artificial-intelligence assistance disclosure. OpenAI Codex, operating with a GPT-5-family model whose exact deployed snapshot identifier was not exposed in the session, provided substantive assistance in August 2026. Under human direction it helped audit repository artifacts; draft and review Python, L^AT_EX, and release files; implement statistical checks and visualizations; verify bibliographic metadata; and draft and edit prose.

Bounded Codex worker agents performed evidence and literature audits. Every admitted number was required to trace to retained raw evidence or a frozen new protocol; outputs were checked through hashes, executable regeneration, tests, and source inspection. The final package must also pass compilation, full-page rendering, and visual review before submission. No generative-image model was used. The human authors remain responsible for all content and must add any substantive earlier AI use not recoverable from repository history. AI systems are not authors.

AUTHOR CONTRIBUTIONS

CRedit roles for conceptualization, methodology, software, validation, formal analysis, investigation, data curation, visualization, writing, supervision, administration, and funding acquisition require author-confirmed names. They are intentionally not inferred from Git history.

COMPETING INTERESTS

No reliable conflict statement was found in the repository. Every author must confirm relevant financial and nonfinancial interests before submission; the corresponding author may state that none exist only after that review.

DATA AVAILABILITY STATEMENT

The intended private-review sanitized companion package is designed to contain processed one-row-per-paired-unit outcomes, statistical summaries, figure inputs, scripts, protocols, and hashes. Its verification code is intended to recalculate primary marginal counts and the equal-weight point estimate from released paired outcomes; its generation scripts are intended to recreate the manuscript tables and figures from those outcomes and released summaries. It does not independently recompute every interval or statistic from engine trajectories. The tournament engine and source, card database, deck files, private replay observations, and third-party opponent packages are not redistributed because of organizer terms, third-party rights, and privacy constraints. Exact gameplay regeneration requires separately authorized organizer access to the engine build identified by hash. The retained feature corpus and policy packages are local ignored artifacts; their hashes, counts, transformations, and aggregates are reported, but they are excluded from the sanitized release pending rights review. Repository DOI, archival location, maintainer contact, and any controlled-access process must be supplied before submission. A reference-list citation for the released

data/software package, including creators, title, year, version, repository, and persistent identifier, must then be added and cited in this statement; those facts are not inferred here.

Appendix A: Encoder and corpus audit

The audit streams the retained gzip corpus, identifies ordinary PLAY options by option type and absence of an explicit raw area, and constructs an exact blind signature from option type, context, zero source, target, attack, area, in-play-area, and numeric fields. Signatures are grouped only within one decision row, because rows have different shared state vectors. A cross-identity collision group must contain at least two simultaneous options with different positive repaired identities. The retained corpus contained no such group. An earlier internal calculation grouped signatures across unrelated observations and was rejected during red-team review; its 17 repeated patterns and 99.7% instance coverage are not collision estimates. Counts are exact over decoded rows. The source-frequency table replaces identifiers with stable anonymous ranks. The corpus hash is `a3d28e9c...dcdc40`; manifest hash is `8c13a5de...1a774`. This audit verifies retained-row behavior but not the provenance of A2's original training corpus.

Appendix B: Training and planned four-cell provenance

The missing C3 cell was generated after its protocol was committed. A first invocation failed at import time before reading or writing data; the protocol records the import-only restart and subsequent path-only repair. A later packaging amendment, still before any C3 gameplay, required a frozen-seed rerun to replace stale copied manifest metadata. That rerun overwrote the first completed C3 package, report, and log. The final package is byte-auditable and the policy-model digest was unchanged, but the overwritten intermediate package cannot be independently audited; only partial session-history provenance survives. The training report field historically named `frozen_protocol_commit` records the execution HEAD, while the initial protocol and its amendments are identified separately in the companion audit. The derived blind gzip uses a fixed timestamp and hash `b0167f91...b111e`. All 37,199 affected fields changed from nonzero to zero and no affected field was already zero. Training completed three epochs; validation selected the best checkpoint under the frozen training function; all frozen parameters were unchanged. Package policy files are byte-identical and the package tree hash is `236afa20...fd54`. Complete histories and hashes are machine-readable.

The subsequent gameplay runs did not satisfy the

frozen repeated-C1 parity gate. The three C1 executions disagreed on 210 of 2,800 seed-condition units, all against the Starmie and Dipplin opponents. This invalidates the planned seven-opponent cross-cell analysis without invalidating the byte audit of C3 training itself. The protocol text remains unchanged as a record of what was planned.

Appendix C: Statistical algorithms

For each primary bootstrap replicate b , let $I_{ci}^{(b)}$ be n_c indices sampled with replacement from cell c . We compute

$$\hat{\Delta}^{(b)} = \frac{1}{14} \sum_c \frac{1}{n_c} \sum_{i \in I_c^{(b)}} D_{ci}, \quad (\text{C1})$$

and take the 2.5 and 97.5 percentiles. McNemar’s exact two-sided value is twice the smaller $\text{Binomial}(b + c, 0.5)$ tail, clipped at one, where b and c are the two discordant directions. Holm adjustment orders the 14 cell values and enforces monotone step-down adjusted values. The planned ablation would have resampled matched four-cell units, but that procedure was not admitted because the repeated-C1 parity gate failed. Any reported within-run or post hoc five-opponent exploratory summaries use their explicitly labeled analysis units and cannot be interpreted as the frozen four-cell estimator. Replay boot-

straps sample episode identifiers; the sequence-oracle interval samples roots. Seeds, replicate counts, and draw handling are fixed in the protocols and summary JSON.

Appendix D: Evidence exclusions and reproducibility boundary

Surviving historical files contain 5,500 nominal direct pair records, including 600 exact overlapping B0 pairs. One proposed reconstruction of four favorable runs has exactly 3,600 pairs but does not reproduce the reported historical aggregate. The seven-policy field JSON, aggregation code, and `meta_weights.json` do not survive. Historical CERT-B rows do not survive, preventing reconstruction of its clustered interval. These values are recorded in the ledger but excluded from evidentiary results.

The branch requested as `grim-5k-variance-floor` existed locally only as `archive/grim-5k-variance-floor`. Large packages and engines are ignored files fixed by hash rather than Git objects. The intended release is designed to support regeneration of manuscript presentation artifacts from processed results and a bounded primary point-estimate check, not full independent statistical or engine replication. Engine-level trajectory reproduction remains conditional on lawful third-party access and cannot be guaranteed by the authors’ package.

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